Announcements and reminders

- **Final exam: Monday, 27 Apr, 5pm-7pm, venue TBA**

We'll use Examplify, and the exam will be open book meaning you may bring one sheet of paper (handwritten or printed, A4 or smaller, double-sided).

The final will be similar to the midterm in terms of question style and difficulty (just more questions). However, there will be differences in administrative matters because the final exam is more strictly regulated by the university. For example, there will be pre-assigned seating and ID checks.

The final may contain some questions that *combine* Weeks 1-5 materials with Weeks 6-13 materials.

- **There will be no lecture next week (Good Friday).**

  As for when your tutorial will be cancelled, please contact your TA.

Can an AI be conscious?

What the Chinese room thought experiment does show

The Chinese room thought experiment shows, assuming the Chinese room is not conscious, that being indistinguishable from a conscious being from the outside, in terms of input and output, is **not sufficient** for being a conscious being.

Showing this, however, falls short of showing that AIs can never be conscious because there might be something else that is sufficient for consciousness that AIs can satisfy.

What could such a sufficient condition for consciousness be?

Can an AI be conscious?

Now, let's turn to the position that

AIs, at least some AIs, can be conscious.

One way to defend this position is to find a **sufficient** condition for consciousness and to show that at least some AIs can satisfy it.

One proposed sufficient condition is having exactly the same **functional organization** (the abstract pattern of causal interaction between various parts of a system, and between these parts and external inputs and outputs) as a conscious system, such as the human brain.

Not only indistinguishable from a human in terms of inputs and outputs (like the Chinese room) but also the inner "program" is the same, although it is implemented in a different material (e.g., silicon).

Two questions: Can we build an AI that satisfies this condition—an isomorph of the human brain?

Is the condition really sufficient for consciousness?

Can an AI be conscious?

Now, let's turn to the position that

AIs, at least some AIs, can be conscious.

"One reason to worry that it might not be possible is that conscious experience might depend on quantum mechanical features of the brain. If it does, science may forever lack the requisite information about your brain to construct a true quantum duplicate of you, because quantum restrictions involving the measurement of particles may disallow learning the precise features of the brain that are needed to construct a true isomorph of you." (p.29)

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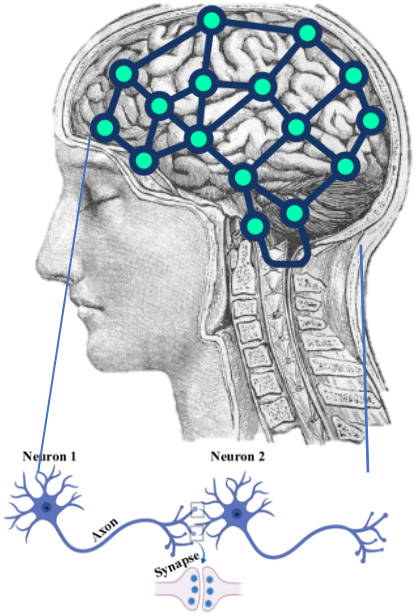
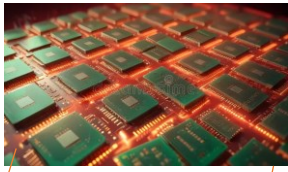
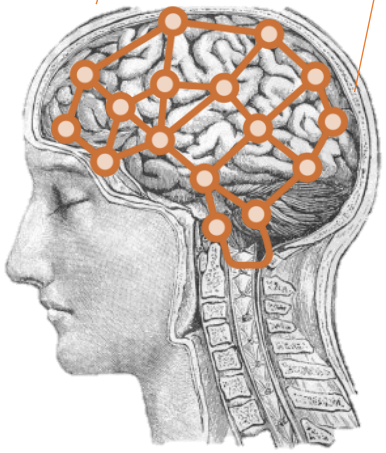
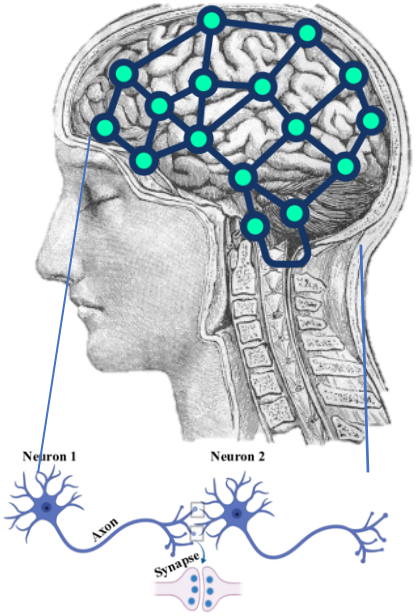


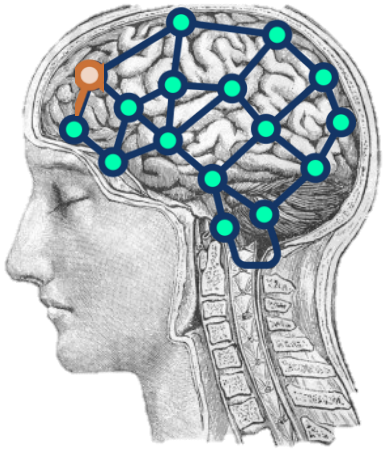
Imagine that you go through a surgery that
gradually replaces your neurons with microchips, one by one.
Each chip performs precisely the same local function as the neuron it replaces.

(We're also assuming that you are not anesthetized and
that you can talk with the doctor during the procedure.)

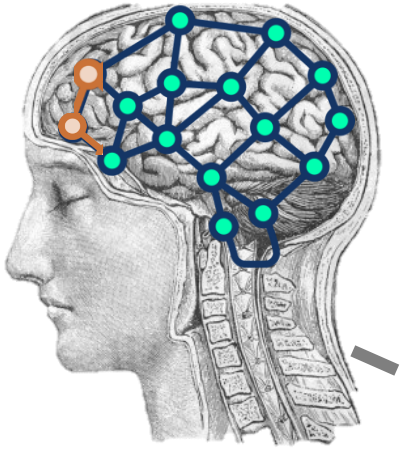








an isomorph



an isomorph

an isomorph

There are three hypotheses about what happens to consciousness during this surgery.

1. The final isomorph is **conscious**.

2. The final isomorph is **not conscious**.
   2-1. Consciousness suddenly disappears at some point during the procedure.
   2-2. Consciousness gradually fades during the procedure.

According to David Chalmers, who proposed this thought experiment, 1 is most plausible.

2-1 is implausible because you go from full consciousness to no consciousness when a single neuron was replaced. This would be a bizarre discontinuity, unlike any other that we find in nature.

2-2 is also implausible given that your brain maintains the same functional organization throughout the process. This means, for example, that if the doctor asks you if you're still fully conscious, you'll judge and say that you are, throughout the process. So, you're completely out of touch with your own consciousness: you think it's normal when in fact it is fading away. This also seems quite bizarre.

- I’ll leave it up to you to think about whether this thought experiment is convincing.

- Note that this thought experiment, even if successful, doesn’t really tell us whether the AIs that are ***not*** isomorphs of the human brain can be conscious or not.

Sufficient condition

X is (a) sufficient (condition) for Y

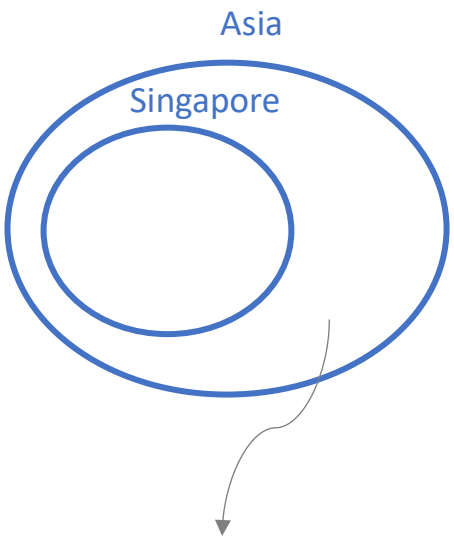
= X is a subset of Y

= X 100% guarantees Y

= If X is the case, Y is the case (always).

= It is not possible for something to be or do X and not Y.

E.g.) Being in Singapore is a sufficient condition for being in Asia.



Notice that being in Singapore is not the **only** way to be in Asia

X being sufficient for Y doesn't mean X is the only way to get Y. X can be sufficient but not necessary for Y.

- I'll leave it up to you to think about whether this thought experiment is convincing.

- Note that this thought experiment, even if successful, doesn't really tell us whether the AIs that are ***not*** isomorphs of the human brain can be conscious or not.

- So, the practical significance of this thought experiment is limited, given that many AIs that are already made or will be made in the near future are likely not isomorphs of the human brain.

Suppose that the second view about AI consciousness is correct.

Can an AI be conscious?

Now, let's turn to the position that

AIs, at least some AIs, can be conscious.

One way to defend this position is to find a **sufficient** condition for consciousness and to show that at least some AIs can satisfy it.

One proposed sufficient condition is having exactly the same **functional organization** (the abstract pattern of causal interaction between various parts of a system, and between these parts and external inputs and outputs) as a conscious system, such as the human brain.

Not only indistinguishable from a human in terms of inputs and outputs (like the Chinese room) but also the inner "program" is the same, although it is implemented in a different material (e.g., silicon).

Two questions: Can we build an AI that satisfies this condition—an isomorph of the human brain?

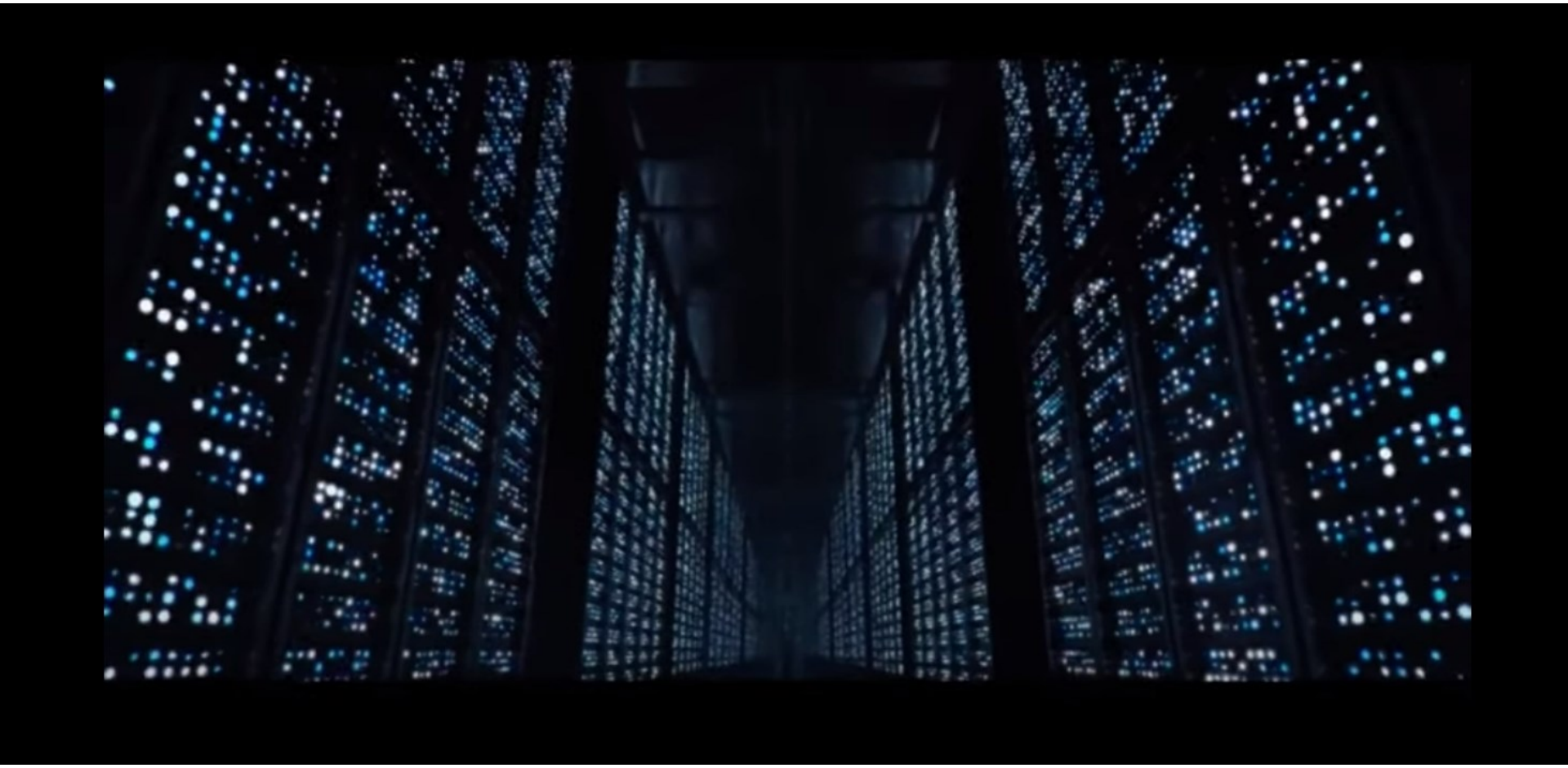
Is the condition really sufficient for consciousness?

Then, consciousness is **substrate-independent**

: consciousness can be implemented not only by carbon-based biological brains

but also by systems made of other substrates/materials.

If so, not only can AIs be conscious, but we could also live in a computer simulation!



A scene from an episode of *Black Mirror*

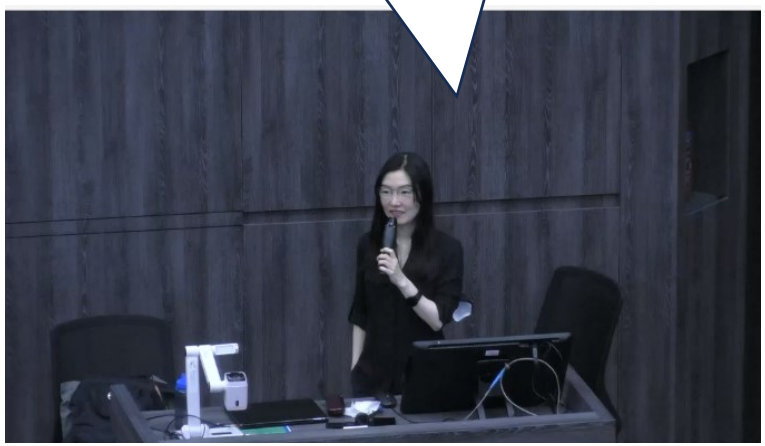
Nick Bostrom, "Do We Live in a Computer Simulation?“ (2006)

Nick Bostrom
(1973-)

We should assign a significant probability to the hypothesis that we’re living in a simulation!

I’m assuming that consciousness is substrate-independent and that it is technologically feasible to create simulations containing minds with experiences like ours.

According to Bostrom,
we should assign a significant
probability to the idea that
we are living in a simulation!

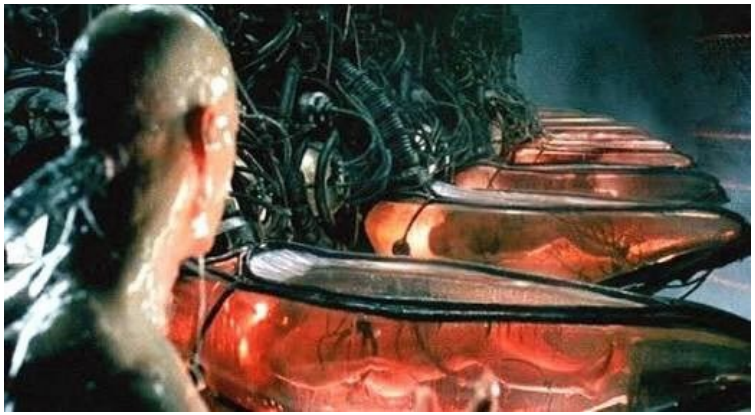


simulation (of 2026)

You ARE living
in a simulation.

Outside the simulation (in 2226, say)

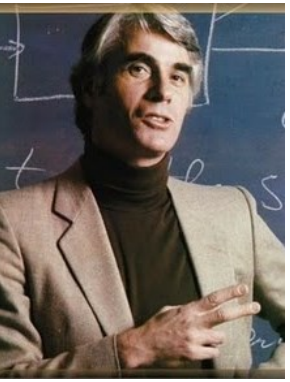
simulation



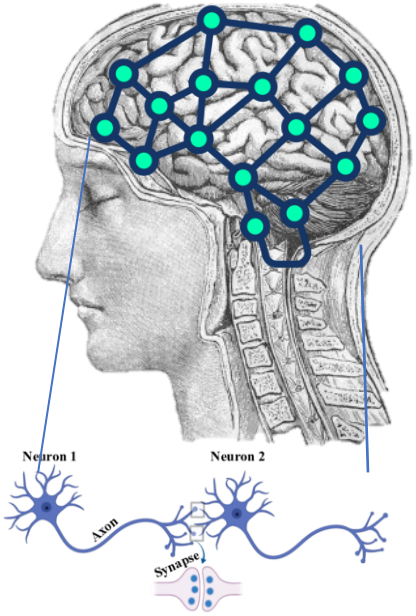
outside the simulation

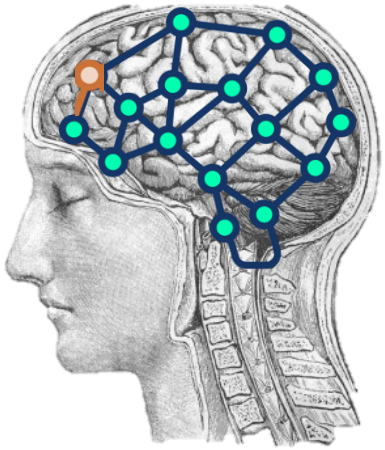
The Experience Machine

"Suppose there were an experience machine that would give you any experience you desired. Superduper neuropsychologists could stimulate your brain so that you would think and feel you were writing a great novel, or making a friend, or reading an interesting book. All the time you would be floating in a tank, with electrodes attached to your brain. Should you plug into this machine for life, preprogramming your life's experiences? If you are worried about missing out on desirable experiences, we can suppose that business enterprises have researched thoroughly the lives of many others. You can pick and choose from their large library or smorgasbord of such experiences, selecting your life's experiences for, say, the next two years. After two years have passed, you will have ten minutes or ten hours out of the tank, to select the experiences o f your next two years. Of course, while in the tank you won't know that you're there; you'll think it's all actually hap pening. Others can also plug in to have the experiences they want, so there's no need to stay unplugged to serve them. (Ignore problems such as who will service the machines if everybody plugs in.) Would you plug in?"

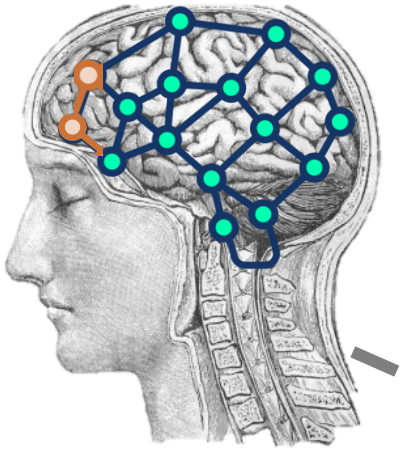


Robert Nozick
(1938 – 2002)



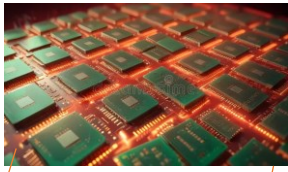


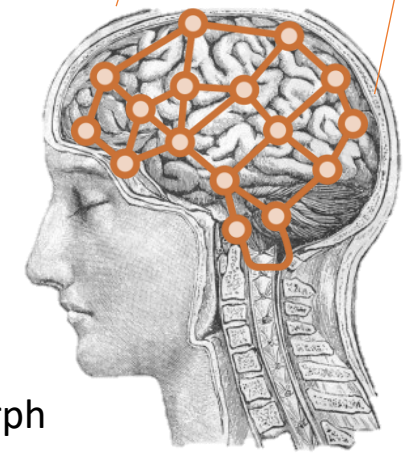
an isomorph



an isomorph

You still have your biological body
(except for the brain).





an isomorph

≈

≈

simulation (of 2026)

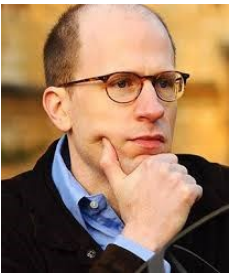
Outside the simulation (in 2226, say)

You ARE living
in a simulation.

You, too,
actually.

If there's no more level of simulation,
this is 'the "bottom" level of reality', which is not simulated.

Why should we assign a significant probability to the hypothesis that we're living in a simulation now?



Nick Bostrom
(1973-)

At least one of the following three possibilities is true:

(a) "Almost all civilisations at our level of development become extinct before becoming technologically mature" 1/3

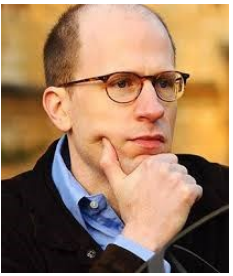
(b) "The fraction of technologically mature civilisations that are interested in creating ancestor simulations is almost zero" 1/3

(c) "You are almost certainly living in a computer simulation" 1/3

This is what he himself calls the conclusion of his argument.

But he also takes a further step.

"In reality, we don't have much specific information to tell us which of the three possibilities might be true. In this situation, it might be reasonable to distribute our credence roughly evenly between them" (p.9)



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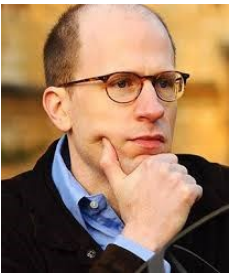
A technologically mature civilization has acquired most of the technological capabilities that are consistent with physical laws and with material and energy constraints.



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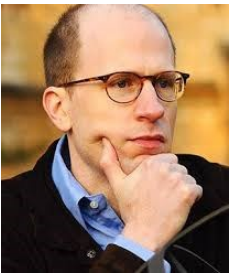
"For example, maybe there is some technology that every advanced civilisation eventually develops and which then destroys them." (p.9)



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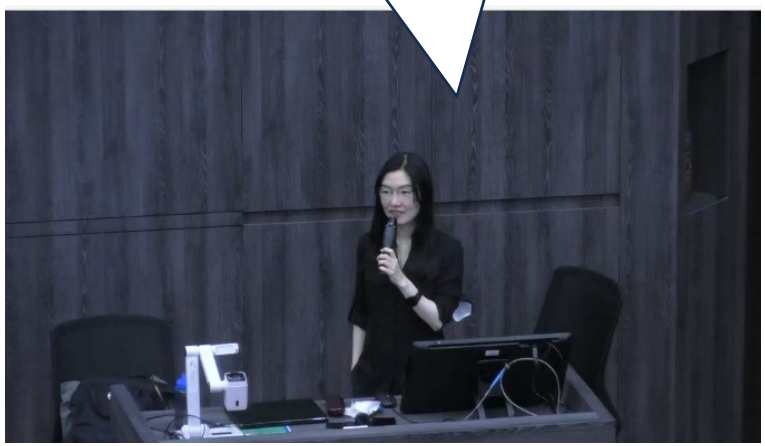
"detailed simulations of the simulators' predecessors – detailed enough for the simulated minds to be conscious and have the same kinds of experiences we have." (p.8)

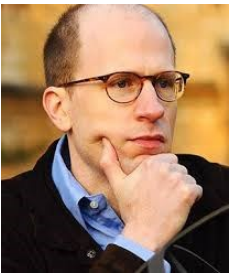
According to Bostrom,
we should assign a significant
probability to the idea that
we are living in a simulation!



simulation (of 2026)

You ARE living
in a simulation.

Outside the simulation (in 2226, say)



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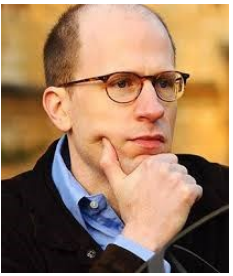
Perhaps because of ethical concerns or simply because they’re not interested.



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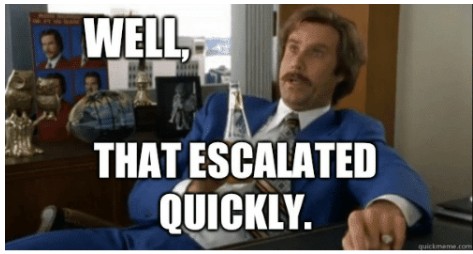
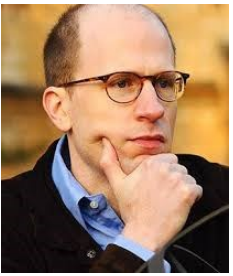
"if you think that possibilities one and two are both false, you should accept the third" (p.9)

What directly follows if both (a) and (b) are false is that a non-negligible fraction of civilisations run ancestor simulations containing minds with experiences like ours.

How does Bostrom go from this to (c)?





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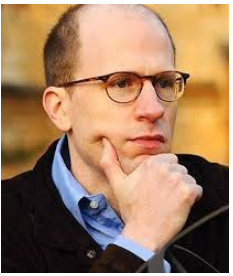
How does Bostrom go from this to (c)?

First, he thinks that civilisations that run such simulations will run lots of them.

"We assume that technologically mature civilisations would have access to enormous amounts of computing power. So enormous, in fact, that by devoting even a tiny fraction to ancestor simulations, they would be able to implement billions of simulations, each containing as many people as have ever existed." (p.9)



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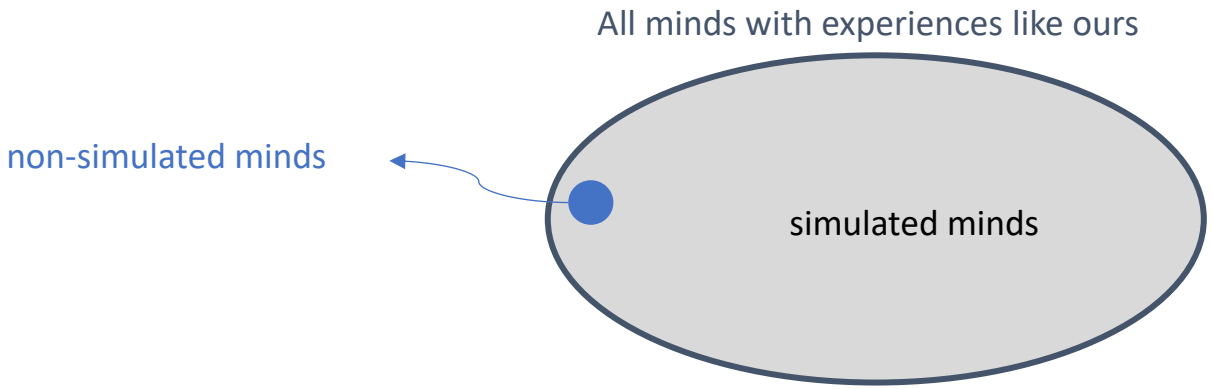
He then draws (c) from this point using the following reasoning…

If there are lots of simulations containing minds with experiences like ours…

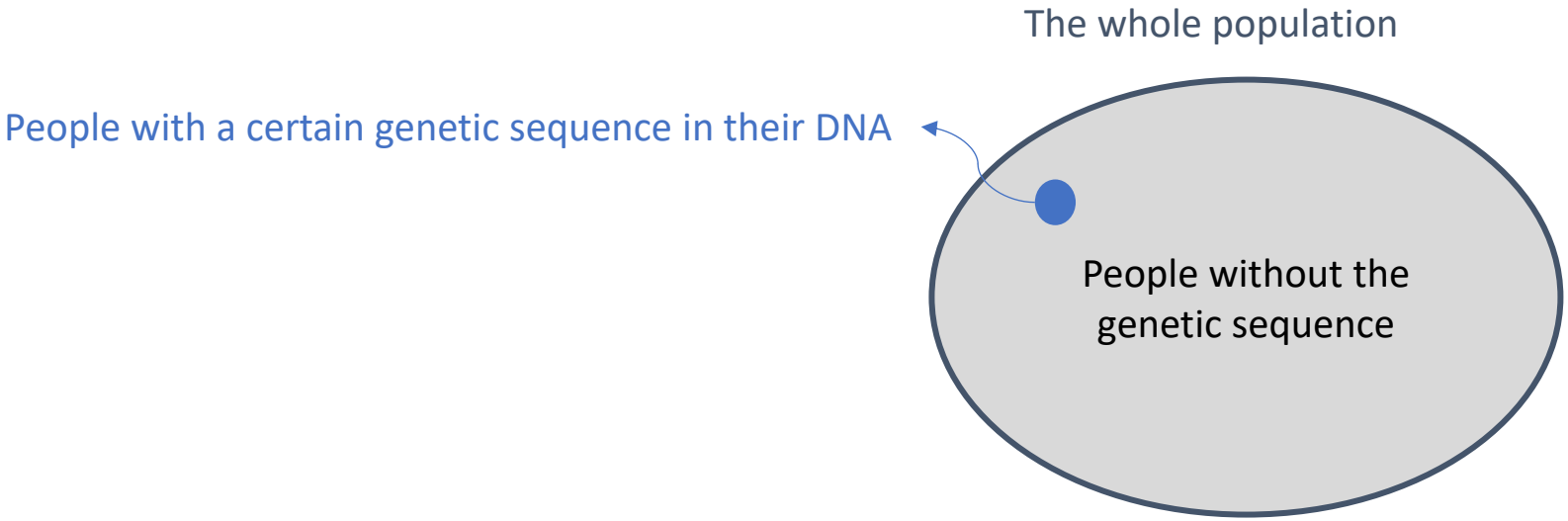


Nick Bostrom
(1973-)

" If we work out the numbers, we find that there would be vastly many more simulated minds than nonsimulated minds. We assume that technologically mature civilisations would have access to enormous amounts of computing power. So enormous, in fact, that by devoting even a tiny fraction to ancestor simulations, they would be able to implement billions of simulations, each containing as many people as have ever existed. In other words, almost all minds like yours would be simulated. Therefore,…, you would have to assume that you are probably one of these simulated minds rather than one of the ones that are not simulated." (p.9)

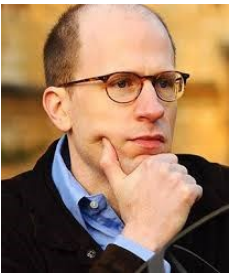
assuming that we have no information that enables us to distinguish between the experiences of simulated minds and non-simulated minds

An example that illustrates how this reasoning works:



Assuming that the genetic sequence is not correlated with any observable characteristics,

you should think that you almost certainly don't have it (unless you have had your DNA sequenced).



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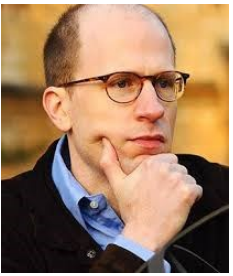
(c) "You are almost certainly living in a computer simulation" 1/3

"if you think that possibilities one and two are both false, you should accept the third" (p.9)

"In reality, we don't have much specific information to tell us which of the three possibilities might be true. In this situation, it might be reasonable to distribute our credence roughly evenly between them" (p.9)

Bostrom might be wrong about this claim. But note that we should still assign a significant probability to (c), insofar as we think that there is a significant probability that both (a) and (b) are false.



Nick Bostrom
(1973-)

At least one of the following three possibilities is true:

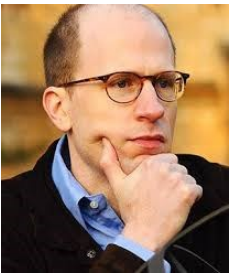
(a) “Almost all civilisations at our level of development become extinct before becoming technologically mature” 1/3

(b) “The fraction of technologically mature civilisations that are interested in creating ancestor simulations is almost zero” 1/3

(c) “You are almost certainly living in a computer simulation” 1/3

<reasons to think that (a) might be false>

- There may be many other civilizations like ours in the universe, and perhaps many of them reach technological maturity.



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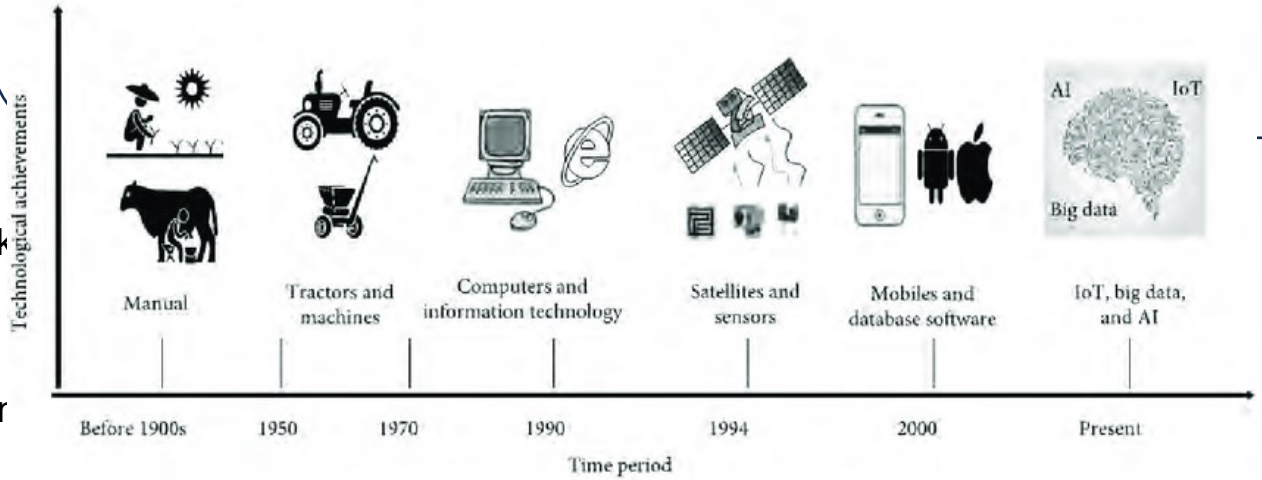
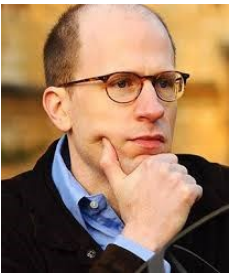


<reasons to think

- There may be reach technological r

- Our continued technological progress gives some reason to think civilizations like ours can keep advancing rather than collapsing early.



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(c) "You are almost certainly living in a computer simulation" 1/3

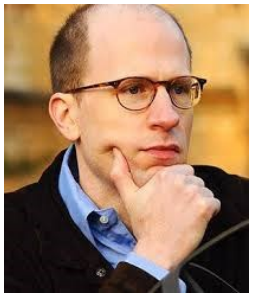
<reasons to think that (b) might be false>

- (b) requires an extreme convergence in ethics, law, and motivation across advanced civilizations. That does not seem obviously likely.

- It might take only one individual in a civilisation to run lots of ancestor simulations.

https://www.youtube.com/watch?v=2KK_kzrJPS8

What kinds of evidence would be relevant to which of the three possibilities is true?



Nick Bostrom
(1973-)

"If we are in a simulation, could [we] ever know for certain? If the simulators don't want us to find out, we probably never will. But if they choose to reveal themselves, they could certainly do so. Another event that would let us conclude with a high degree of confidence that we are in a simulation is if we ever reach a point when we are about to switch on our own ancestor simulations. That would be very strong evidence against the first two propositions, leaving us only with the third.(p.9)

I’ll leave it up to you to think about whether Bostrom’s reasoning is convincing and, if it isn’t, where it goes wrong.

Another thing you might want to think about:

What implications would follow if we're indeed living in a simulation?

Nick Bostrom
(1973-)

"Your first reaction might think that if three is true, then all bets are off and you would go crazy. To reason thus would be an error. Even if we are in a simulation, the best methods of predicting what will happen next are still the familiar ones – extrapolation of past trends, scientific modelling and common sense. To a first approximation, if you thought you were in a simulation, you should get on with your life in much the same way as if you were convinced that you were leading a non-simulated life at the "bottom" level of reality." (p.9)

More questions you might want to think about:

Is it possible to live a good life in a simulation?

Can we have free will in a simulation?

What to do this week

- **No lecture next week**

- Do the reading for next lecture

: Levinson, "Defining Art Historically"